

SAMRUDDHI MAGDUM

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PROFESSIONAL SUMMARY

AI Developer with 11 months of production internship experience building AI-powered applications, backend REST APIs, and ML pipelines using Python and FastAPI. Developed a multi-modal stress and depression detection system integrating ECG, HRV, and PHQ-9 analysis, and an AI test generation platform serving real business clients. Pursuing B.Tech in AI & Machine Learning (90%, graduating 2026). Targeting AI Engineer, ML Engineer, and Python Developer roles.

TECHNICAL SKILLS

Programming: Python | SQL

AI & ML: Machine Learning | Deep Learning | Computer Vision | NLP

Frameworks: FastAPI | TensorFlow | Scikit-learn | OpenCV

GenAI: Prompt Engineering

Databases: PostgreSQL | MySQL

Tools: Git | GitHub | Docker

PROFESSIONAL EXPERIENCE

AI Developer Intern | QualTech Soft Pvt. Ltd., Kolhapur | Aug 2025 – Jun 2026

- Built AI-powered backend modules in Python and FastAPI, delivering production features for live business client applications.
- Designed and deployed RESTful APIs connecting FastAPI services with React.js frontends, enabling real-time AI feature delivery.
- Engineered an AI chatbot with iterative prompt engineering and structured testing, improving response coherence and reducing failure rates across client scenarios.
- Led backend development of an AI test generation platform, building APIs for assessment creation, dynamic question management, and automated scoring workflows.

PROJECTS

AI-Based Stress & Depression Detection System | [GitHub \(link\)](#)

Tech Stack: Python | FastAPI | TensorFlow | Scikit-learn (SVM) | React.js | PostgreSQL

- Built a multi-modal ML web application assessing stress and depression risk by analyzing ECG signals, HRV features (18 extracted features), lifestyle inputs, and PHQ-9 questionnaire responses.
- Trained an SVM classifier achieving 98.75% test accuracy, 0.9982 ROC-AUC, and F1-score of 0.9874 on a 240-sample stress classification task (WESAD dataset integration in progress).
- Developed a full ML pipeline covering data preprocessing, feature extraction, model inference, and automated PDF report generation via FastAPI.

AI Test Generation Assistant | [GitHub \(link\)](#)

Tech Stack: Python | FastAPI | PostgreSQL | React.js

- Developed an AI-assisted platform for creating and managing psychological and educational assessments, deployed for a live business client.
- Designed RESTful backend APIs managing test types, dimensions, question banks, and end-to-end assessment workflows.
- Built a dynamic React.js interface enabling non-technical users to configure and publish assessments, significantly reducing manual creation effort.

EDUCATION

B.Tech in Artificial Intelligence & Machine Learning — D.Y. Patil ATU | Aug 2023 – Sep 2026 | **90%**

Diploma in Computer Engineering — Datta Polytechnic, Shirol | Jul 2020 – Jun 2023 | **71.89%**